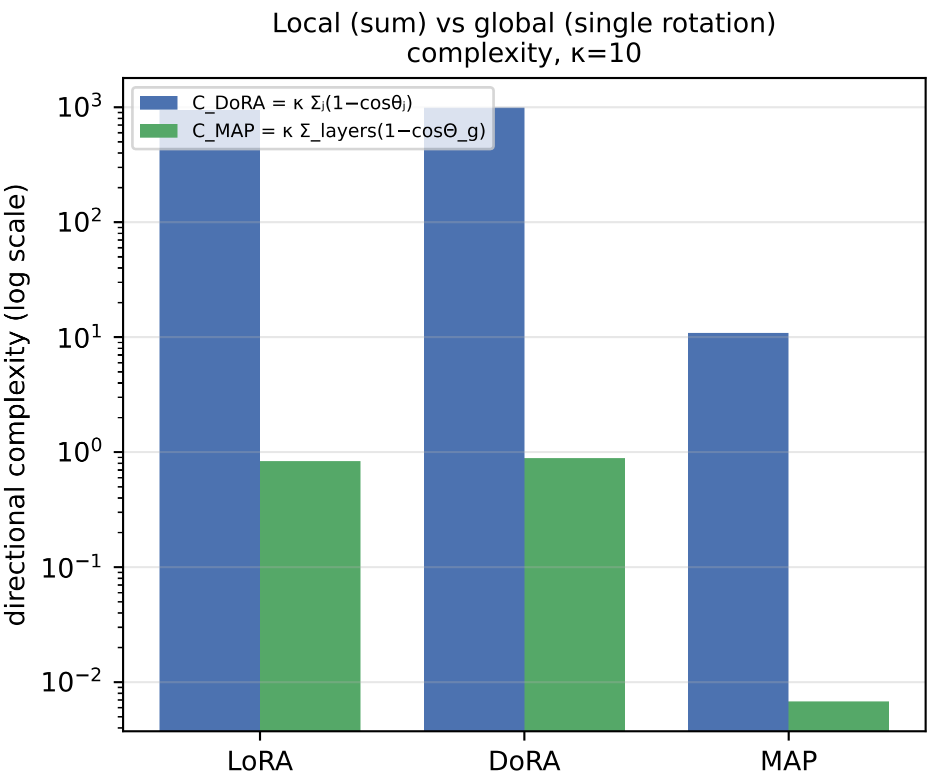
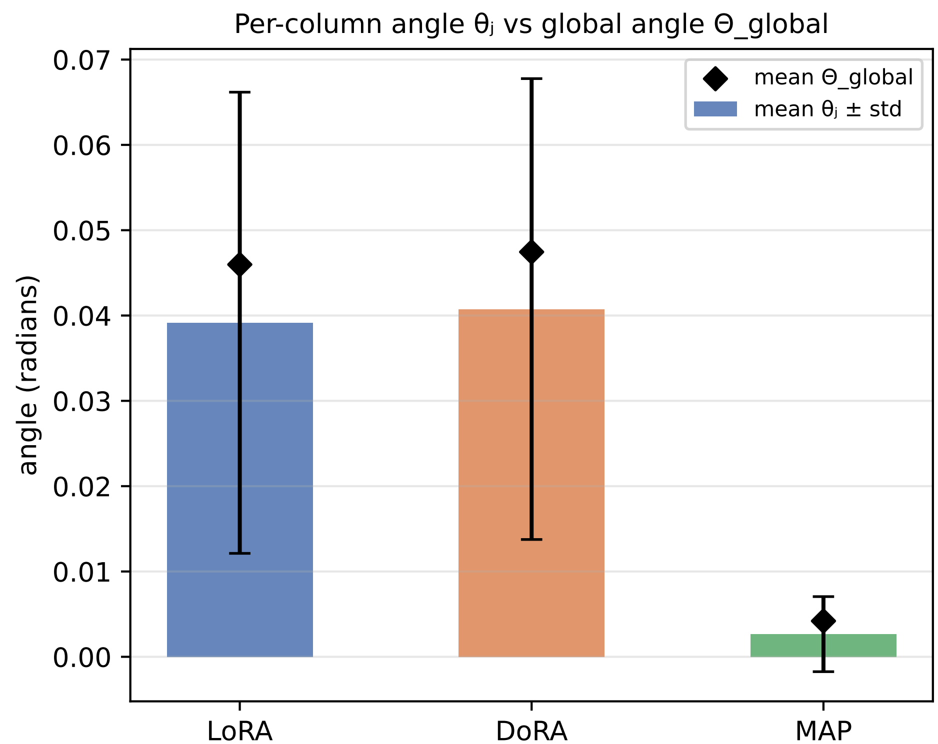
Directional Updates: LoRA vs DoRA vs MAP on RoBERTa-base / QNLI

Fine-tuning metrics and aggregate directional complexity

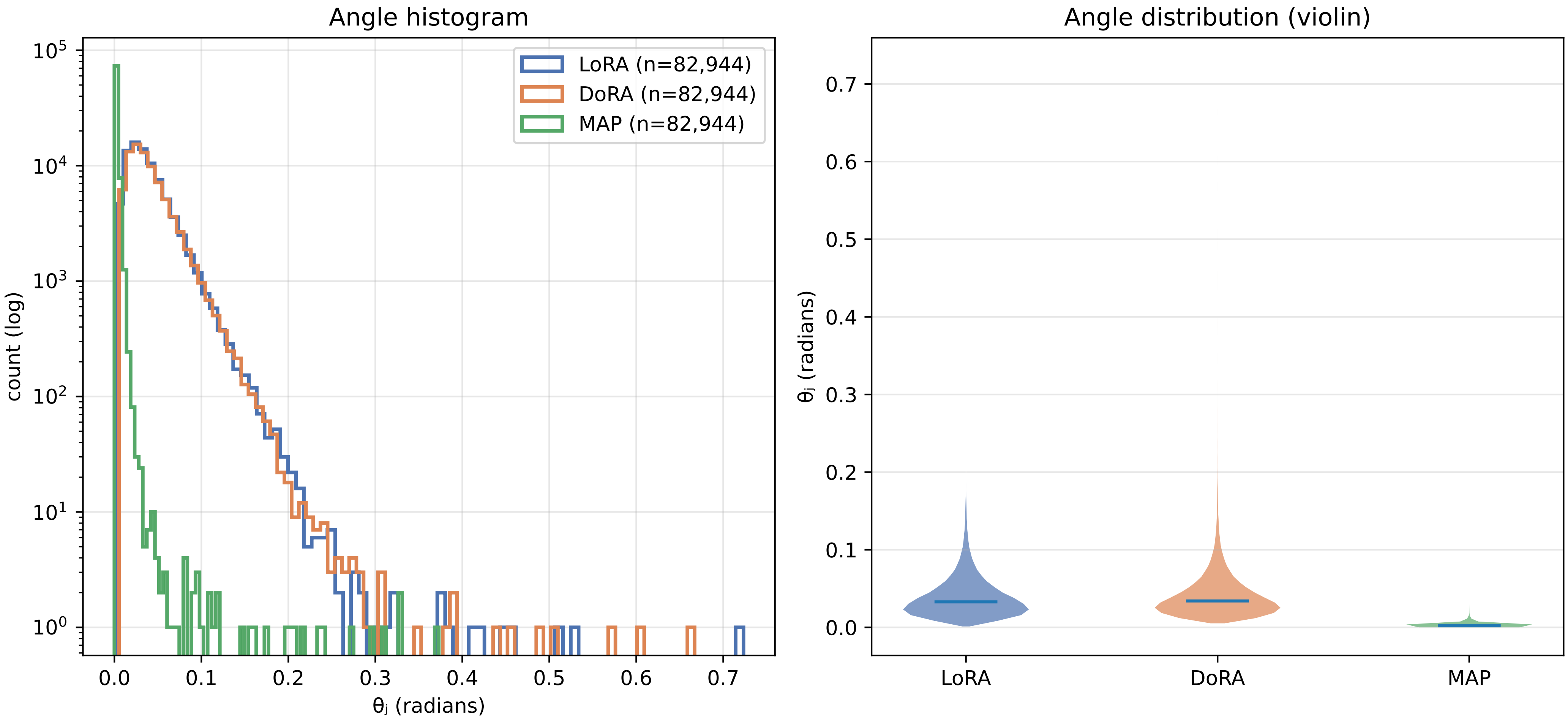
Method	Eval acc	Trainable params	C_DoRA (Σ local)	C_MAP (global)	mean θ_j (rad)	mean Θ_g (rad)	mean s/d
LoRA	92.64%	1,919,234	937.69	0.832	0.0392	0.0460	0.969
DoRA	93.12%	2,002,178	990.49	0.878	0.0408	0.0475	0.988
MAP	89.51%	1,919,378	10.92	0.007	0.0027	0.0042	0.022



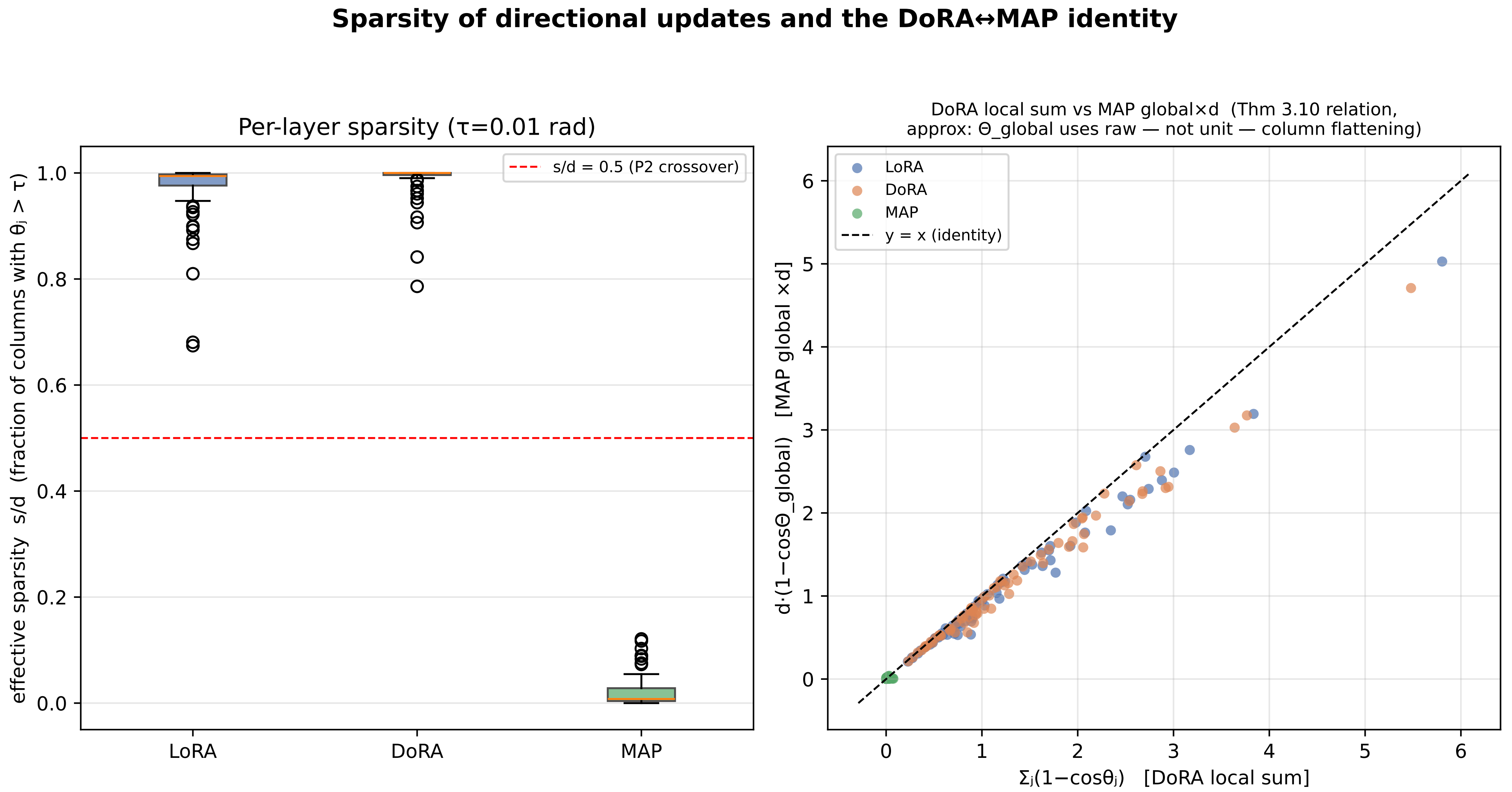


Θ_global is the single rotation of the whole flattened matrix (MAP view); θ_j are the per-column rotations (DoRA view).
Exact identity (Thm 3.10): $1-\cos\Theta\_global = mean\_j(1-\cos\theta_j)$ → DoRA sums what MAP averages.

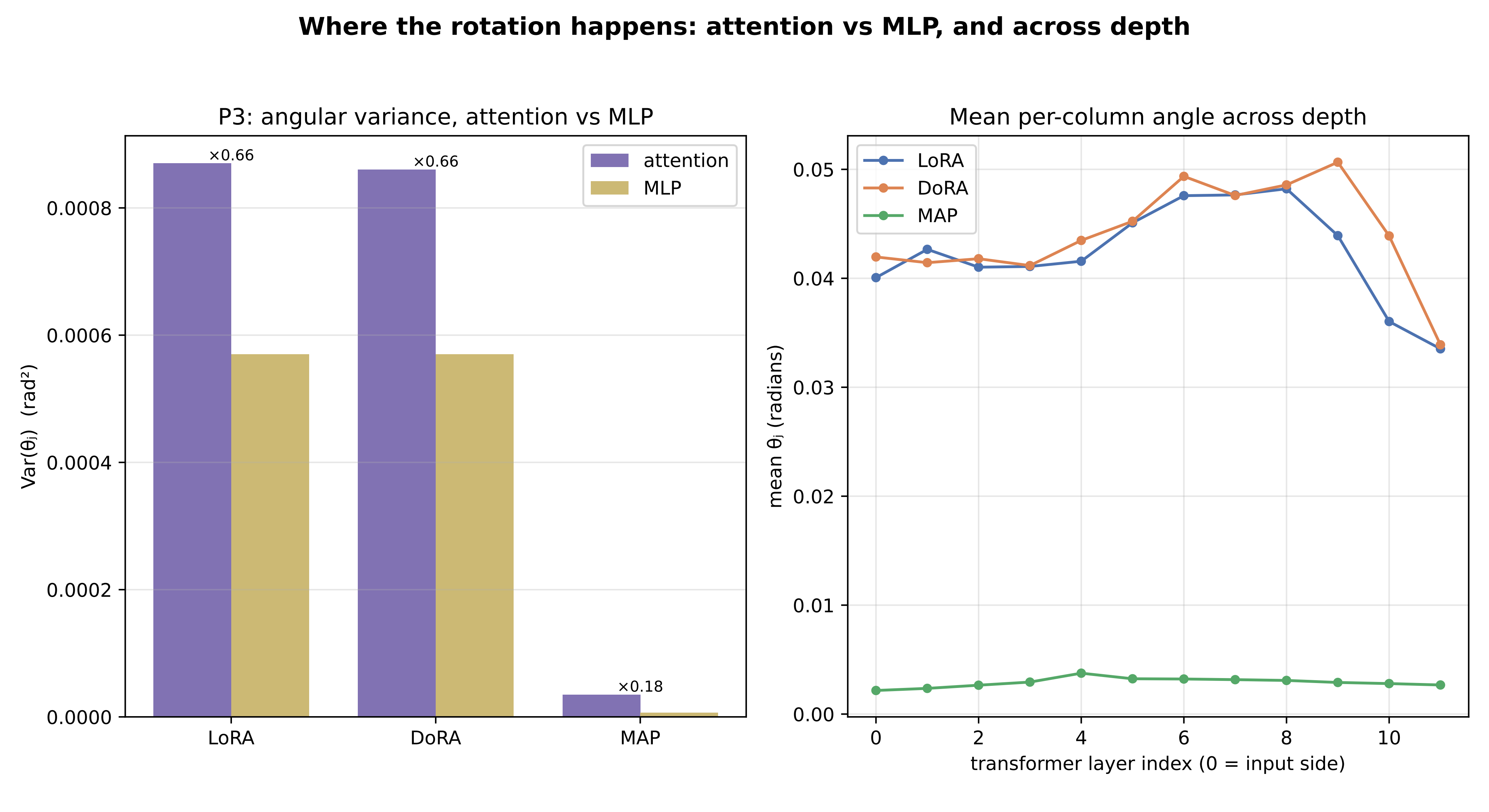
Distribution of per-column directional change θ_j (pooled over all target layers)



A long right tail = a few columns rotate a lot (sparse/localized update). A concentrated body near 0 = most columns barely move.



Sparsity of directional updates and the DoRA↔MAP identity
Per-layer sparsity (τ=0.01 rad)
effective sparsity s/d (fraction of columns with θⱼ > τ)
s/d = 0.5 (P2 crossover)
LoRA
DoRA
MAP
DoRA local sum vs MAP global×d (Thm 3.10 relation,
approx: Θ_global uses raw — not unit — column flattening)
LoRA
DoRA
MAP
y = x (identity)
d·(1−cosΘ_global) [MAP global ×d]
Σⱼ(1−cosθⱼ) [DoRA local sum]



Where the rotation happens: attention vs MLP, and across depth
P3: angular variance, attention vs MLP
×0.66
×0.66
×0.18
attention
MLP
Var(θj) (rad²)
0.0000
0.0002
0.0004
0.0006
0.0008
LoRA
DoRA
MAP
Mean per-column angle across depth
LoRA
DoRA
MAP
mean θj (radians)
0.00
0.01
0.02
0.03
0.04
0.05
0
2
4
6
8
10
transformer layer index (0 = input side)